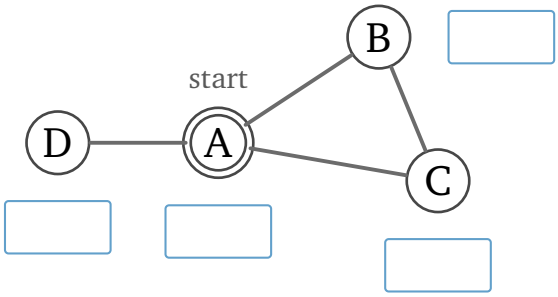
Part 1 — A walker with no map

Four friends: A, B, C and D. A line joins two people who are friends.

A walker starts at A. At every step it picks one friend of the person it is standing on, each friend with the same chance, and moves there. It has no map and no memory of where it came from.



Question 1: Before you compute anything, guess. After a very long walk, at which of the four people is the walker found most often? Why?

Question 2: In each cell of the table, write the chance that the walker moves from the person of that row to the person of that column in one step.

from \ to	**A**	**B**	**C**	**D**
A				
B				
C				
D				

Then add up each row. What do you get?

Question 3: The walker starts at A. What is the chance that it is at B after exactly three steps?

A route is the list of people the walker stands on, for example A → D → A → B. The chance of a route is the chances of its three steps multiplied together: $\frac{1}{3} \times 1 \times \frac{1}{3} = \frac{1}{9}$.

Write every three-step route from A to B in the table, with its chance. Then add up the chances.

route	**chance**
A → D → A → B	$\frac{1}{3} \times 1 \times \frac{1}{3} = \frac{1}{9}$

Part 2 — Drunk at the karate club

Now a real network: a university karate club in the 1970s, with 34 members and 78 friendships between them. The same walker walks on it.

Open the simulator: point a camera at the square, or type `go.skojaku.com/m07sim`.
On the screen, a member is a **node** and a friendship is an **edge**.

Question 4: Press **Run 5,000 steps** four times: 20,000 steps.

Under the buttons, two boxes name the node with the most visits and the node with the fewest. Each box gives the node's number of edges and its share of all the visits. Write them in the table.

Then divide the edges by the share.

	node	**edges**	**share**	**edges ÷ share**
most visits				
fewest visits				

Question 5: Look at the two numbers in the last column of the table. Are they the same? Compare them with your neighbour's.

The club has 78 friendships. How are your numbers related to 78?

Question 6: Go back to the four friends in Part 1. They have 4 friendships.

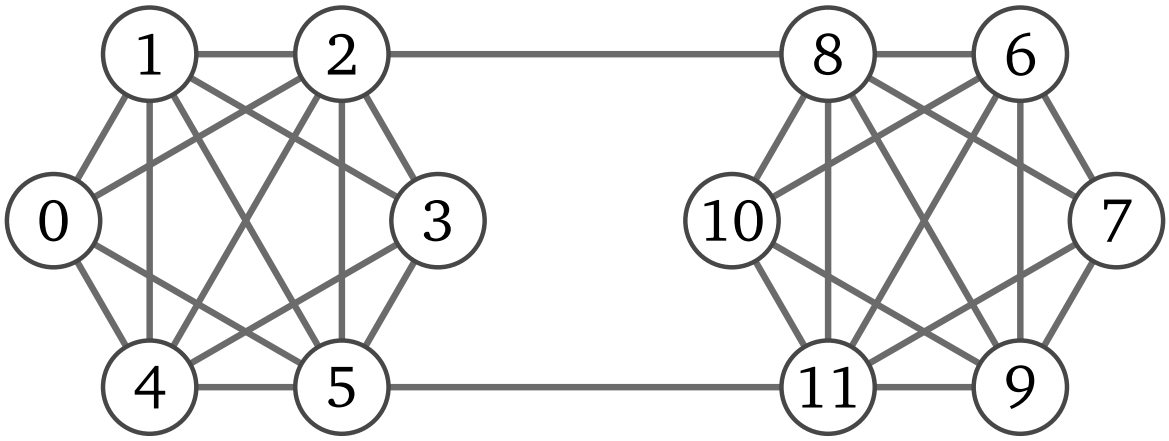
Use what you found in Question 5 to work out the share of time the walker spends at A, B, C and D after a very long walk. Write the four shares in the boxes on the drawing in Part 1.

Was your guess in Question 1 right?

STOP here. Wait for the lecture before you turn the page.

Part 3 — Two rooms

Twelve members and 30 friendships. The walker is the same as in Part 1.

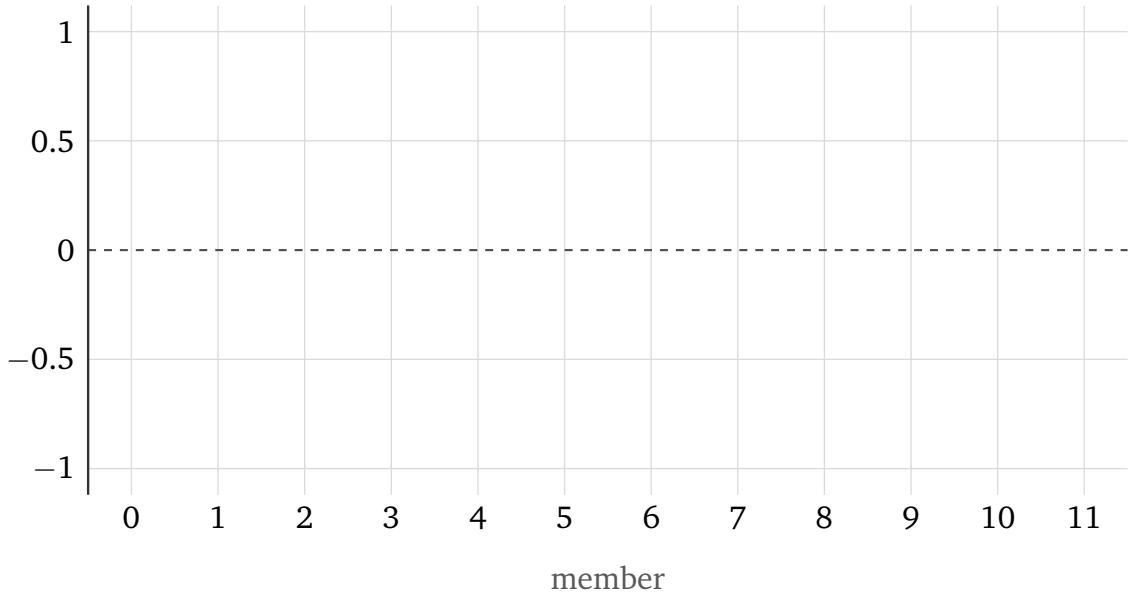


Open the notebook: point a camera at the square, or type `go.skojaku.com/m07lab`. A laptop or a phone will do; if you have neither, pair up.
It has one cell, printed below. Run it. The matrix `P` in it is the table you wrote in Question 2, for these twelve members.

```
import numpy as np

A = np.zeros((12, 12))                          # who is friends with whom
for group in [range(0, 6), range(6, 12)]:
    for i in group:
        for j in group:
            if i != j:
                A[i, j] = 1
for i, j in [(0, 3), (7, 10)]:
    A[i, j] = A[j, i] = 0
for i, j in [(2, 8), (5, 11)]:
    A[i, j] = A[j, i] = 1

P = A / A.sum(axis=1, keepdims=True)            # row i: the chances of leaving i
values, vectors = np.linalg.eig(P)
order = np.argsort(-values.real)                # largest eigenvalue first

print("eigenvalues:", np.round(values.real[order], 3) + 0.0)
v = vectors[:, order[1]].real                   # the second eigenvector
print("second eigenvector:", np.round(v / np.abs(v).max(), 2) + 0.0)
```

Question 7: The cell prints two lines: the twelve eigenvalues of `P`, largest first, and the second eigenvector, one number per member.

Mark the twelve numbers of the second eigenvector as dots on the axes.



Question 8: On the drawing in Part 3, darken every member whose number is above zero. What do the darkened members have in common?

Question 9: Which four members have the numbers closest to zero? Where are they on the drawing?

Question 10: The second eigenvalue is 0.889. Work out 0.889^5 and 0.889^{40}. A phone calculator is fine.

Of the other eigenvalues, the one furthest from zero is -0.423. Work out 0.423^5.

After 5 steps, which eigenvalues still have a term in the walk? And after 40 steps?

Question 11: The lecture asked: short walks show communities, long walks show degree. Why?

Answer in two sentences: one for the long walk, using Part 2, and one for the short walk, using Part 3.

Question 12 (finished early): Members 2 and 8 stop being friends. In the code, delete `(2, 8)`, from the last list and run the cell again.

Is the second eigenvalue now closer to 1, or further from 1?

Does a walker that starts at member 0 stay among members 0 to 5 for longer, or for less long? Why?